LAPORAN JOBSHEET
“2-Setup Project”
MATA KULIAH PEMROGRAMAN BERBASIS FRAMEWORK

M. FATIH AL GHIFARY

2341720194 / TI-3A

PROGRAM STUDI D-IV TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI MALANG

2026

1. Melakukan node -v dan npm -v

```
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework> node -v
v24.12.0
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework> npm -v
11.6.2
```

2. Memberikan perintah git -v untuk melihat versi git

```
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework> git -v
git version 2.43.0.windows.1
```

3. Membuat folder praktikum

```
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1> mkdir praktikum


    Directory: D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/19/2026   1:58 PM                praktikum


PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1> cd .\praktikum\
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1\praktikum>
```

4. Membuat apps next.js

```
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1\praktikum> npx create-next-app@13.4.7
Need to install the following packages:
create-next-app@13.4.7
Ok to proceed? (y) y

√ What is your project named? ... my-app
√ Would you like to use TypeScript with this project? ... No / Yes
√ Would you like to use ESLint with this project? ... No / Yes
√ Would you like to use Tailwind CSS with this project? ... No / Yes
√ Would you like to use `src/` directory with this project? ... No / Yes
√ Use App Router (recommended)? ... No / Yes
√ Would you like to customize the default import alias? ... No / Yes
Creating a new Next.js app in D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1\praktikum\my-app.

Using npm.

Initializing project with template: default-tw


Installing dependencies:
- react
- react-dom
- next
- typescript
- @types/react
- @types/node
- @types/react-dom
- tailwindcss
- postcss
- autoprefixer
- eslint
- eslint-config-next
```

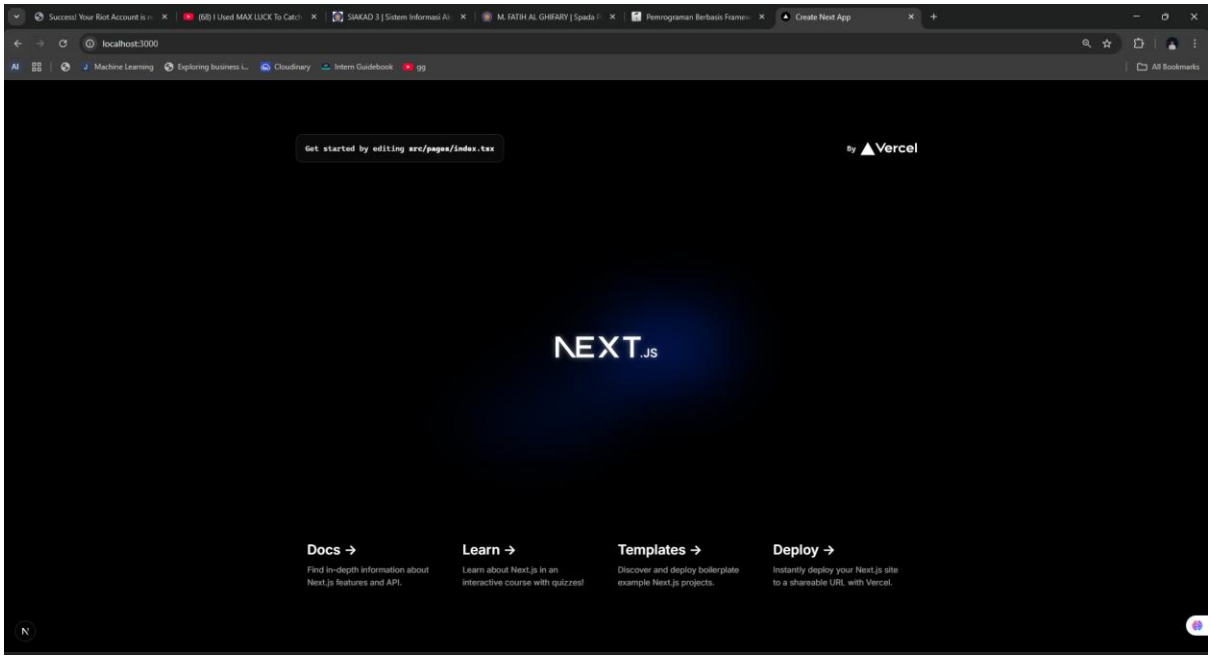
5. Masuk ke folder projectnya

```
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1\praktikum> cd .\my-app\
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1\praktikum\my-app>
```

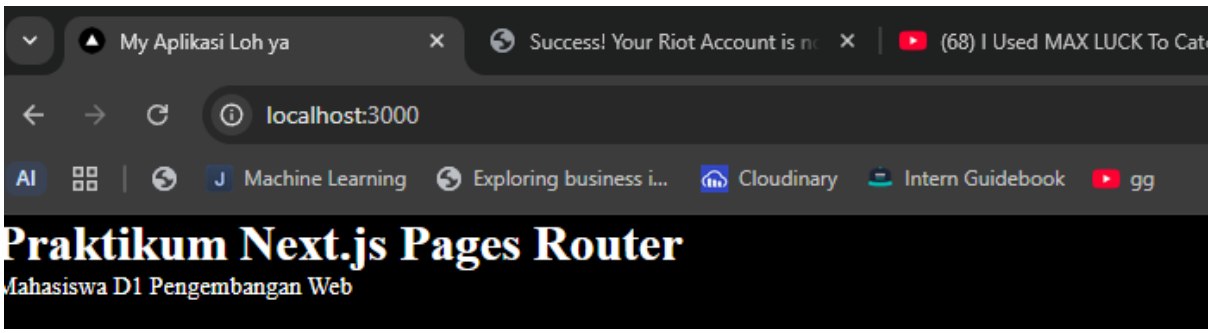
6. Jalankan Aplikasi

```
PS D:\KULIAH\SEMESTER-6\Pemrograman Framework\P1\praktikum\my-app> npm run dev

 > my-app@0.1.0 dev
 > next dev

 ▲ Next.js 16.1.6 (Turbopack)
 - Local:         http://localhost:3000
 - Network:       http://100.102.141.86:3000

 ✓ Starting...
 Attention: Next.js now collects completely anonymous telemetry regarding usage.
 This information is used to shape Next.js' roadmap and prioritize features.
 You can learn more, including how to opt-out if you'd not like to participate in this anonymous program, by visiting the following URL:
 https://nextjs.org/telemetry


   We detected TypeScript in your project and reconfigured your tsconfig.json file for you.
   The following mandatory changes were made to your tsconfig.json:

         - jsx was set to react-jsx (next.js uses the React automatic runtime)

 ✓ Ready in 9.6s
```

7. Buka Localhost



8. Ubah isi Halaman di index.tsx



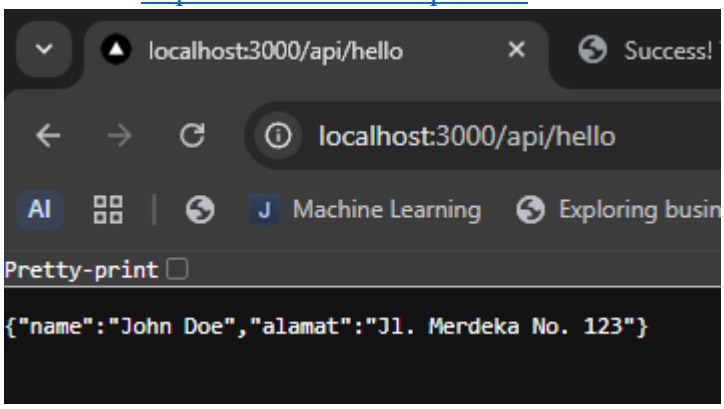
9. Modifikasi API

index.tsx M hello.ts M

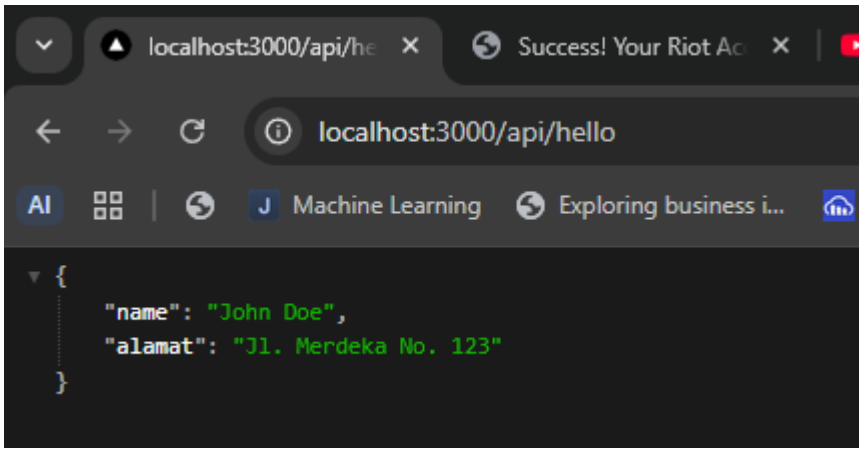
P1 > praktikum > my-app > src > pages > api > hello.ts > handler

```
// Next.js API route support: https://nextjs.org/docs/api-routes/introduction
import type { NextApiRequest, NextApiResponse } from 'next'

type Data = {
  name: string
  alamat: string
}

export default function handler(
  req: NextApiRequest,
  res: NextApiResponse<Data>
) {
  res.status(200).json({ name: 'John Doe', alamat: 'Jl. Merdeka No. 123' })
}
```

10. Membuka http://localhost:3000/api/hello

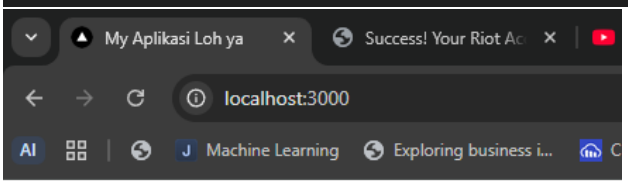


11. Tambah Extension JSON formatter di chrome



12. Ubah file _app.tsx dan buka localhost

```
index.tsx M    hello.ts M    _app.tsx M ×
P1 > praktikum > my-app > src > pages > _app.tsx > ...
1  //import '@/styles/globals.css'
2  import type { AppProps } from 'next/app'
3
4  export default function App({ Component, pageProps }: AppProps) {
5    return <Component {...pageProps} />
6  }
7
```



Praktikum Next.js Pages Router

Mahasiswa D1 Pengembangan Web

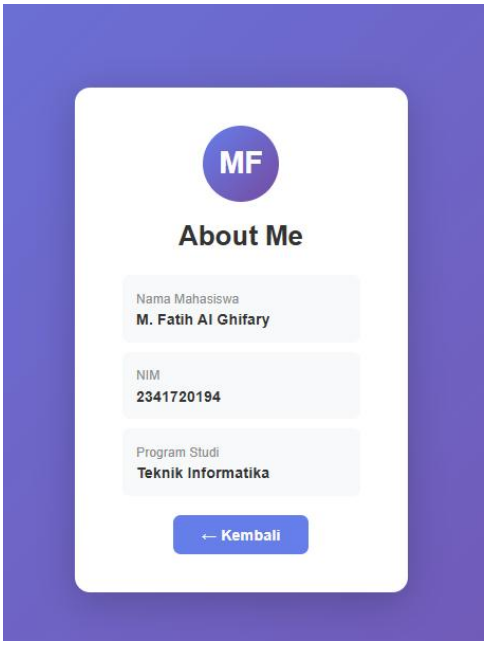
13. Setup ext pada VSCode (sudah terinstall semua)

TUGAS PRAKTIKUM

Tugas 1 (Wajib)

1. Buat halaman baru about.js di folder pages
2. Tampilkan:
   - Nama Mahasiswa
   - NIM
   - Program Studi

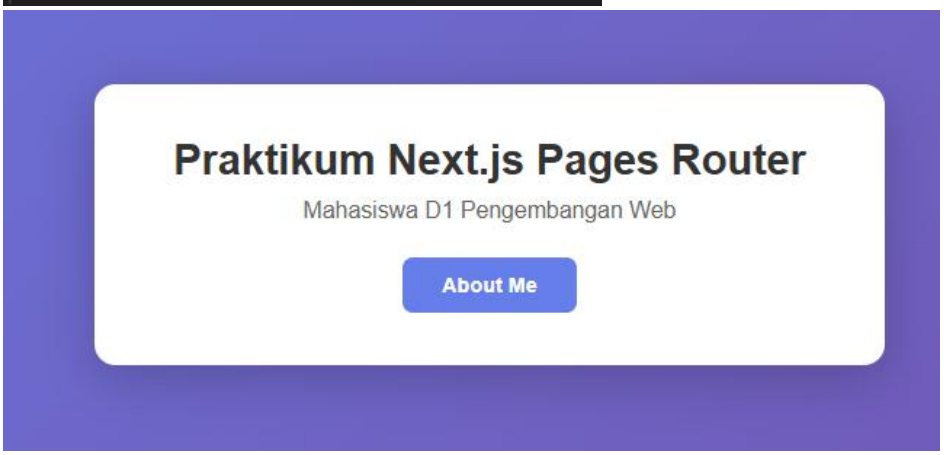
```
<div style={{ textAlign: 'left' }}>
  <div style={{
    padding: '15px',
    backgroundColor: '#f8f9fa',
    borderRadius: '8px',
    marginBottom: '10px'
  }}>
    <span style={{ color: '#888', fontSize: '0.85rem' }}>Nama Mahasiswa</span>
    <p style={{ margin: '5px 0 0', fontWeight: 'bold', color: '#333' }}>M. Fatih Al Ghifary</p>
  </div>
  <div style={{
    padding: '15px',
    backgroundColor: '#f8f9fa',
    borderRadius: '8px',
    marginBottom: '10px'
  }}>
    <span style={{ color: '#888', fontSize: '0.85rem' }}>NIM</span>
    <p style={{ margin: '5px 0 0', fontWeight: 'bold', color: '#333' }}>2341720194</p>
  </div>
  <div style={{
    padding: '15px',
    backgroundColor: '#f8f9fa',
    borderRadius: '8px',
    marginBottom: '20px'
  }}>
    <span style={{ color: '#888', fontSize: '0.85rem' }}>Program Studi</span>
    <p style={{ margin: '5px 0 0', fontWeight: 'bold', color: '#333' }}>Teknik Informatika</p>
  </div>
</div>
```



Tugas 2 (Pengayaan)

1. Tambahkan minimal 1 link navigasi dari halaman utama ke halaman about

```
<Link href="/about" style={{
  backgroundColor: '#667eea',
  color: 'white',
  padding: '12px 30px',
  borderRadius: '8px',
  textDecoration: 'none',
  fontWeight: 'bold',
  display: 'inline-block',
  transition: 'transform 0.2s'
}}>
  About Me
</Link>
```



Pertanyaan Refleksi

1. Mengapa Pages Router disebut sebagai routing berbasis file?
   Jawab: Pages Router pada **Next.js** disebut routing berbasis file karena sistem penentuan URL aplikasi dibuat langsung dari struktur folder dan file di dalam direktori pages. Jadi setiap file yang dibuat otomatis menjadi route tanpa perlu konfigurasi manual seperti di React biasa. Misalnya file about.js akan menghasilkan route /about, sehingga pengaturan navigasi menjadi lebih sederhana dan terstruktur.

2. Apa perbedaan Next.js dengan React standar (CRA)?
   Jawab: Perbedaan utama antara Next.js dan React standar seperti **Create React App** adalah fitur bawaan dan cara rendering. Next.js menyediakan fitur siap pakai seperti server-side rendering (SSR), static site generation (SSG), routing otomatis, dan optimasi performa, sedangkan React standar hanya fokus pada library UI sehingga developer harus menambahkan routing, konfigurasi build, dan optimasi sendiri. Next.js dikembangkan oleh Vercel, sementara React dikembangkan oleh Meta Platforms.

3. Apa fungsi perintah npm run dev?
   Jawab: Perintah npm run dev digunakan untuk menjalankan aplikasi dalam mode development (pengembangan). Mode ini biasanya mengaktifkan server lokal, menampilkan error secara langsung, dan mendukung fitur auto-reload ketika kode diubah sehingga memudahkan proses coding dan testing.

4. Apa perbedaan npm run dev dan run build ?
   Jawab: Npm run dev digunakan saat pengembangan aplikasi dengan fitur debugging dan reload otomatis, sedangkan npm run build digunakan untuk menyiapkan aplikasi versi produksi yang sudah dioptimasi (kode diperkecil, performa ditingkatkan, dan siap di-deploy ke server). Jadi dev untuk proses pengembangan, sedangkan build untuk tahap distribusi atau deployment aplikasi.